

## ▲ Pyramid Problems · First Pyramid — English Worksheet

**Topic:** First Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

The agent **walks the edges** of a square, dropping blocks as it moves. Stack the squares, shrink each one, and a pyramid grows.

### 1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
  turn left
```

What shape does the agent draw? How many blocks long is each side?

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Why is it **repeat 4 times** and **turn left** ? What would **repeat 3 times** draw instead?

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```
set f to 6
repeat 3 times:
  [draw one f × f square]
  move up by 1
  change f by -2
```

List the side length of each layer from bottom to top. Looking from the side, what shape do the stacked squares make?

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```
place on move ON  
move forward by 4  
place on move OFF  
move forward by 4
```

Where do the blocks end up? Why does turning **place on move** OFF matter when the agent moves up to start the next layer?

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## 2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

**Bug A** — The agent is supposed to draw a closed **square** outline.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — Each new layer must be **2 blocks smaller** than the one below, so the tower narrows into a pyramid.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  move up by 1
  change f by 2
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — Each square must sit one block **higher** than the last.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  change f by -2
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Show Your Work 📷 🎥

Switch to your homework world. Run `pyra`, change the **starting value of** `f` to build a shorter pyramid, then reset a stuck agent with `1` (left), `r` (right), `r1` (back).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.